

Foundation Program in Literacy, Numeracy and Skills

NUMERACY
Student's Worksheet
Volume of Cubes and Cuboids

Lesson 8: Volume of a cube

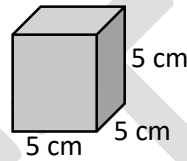
Example

Find the volume of the cube of length of 5 cm.

Solution

Length = 5 cm

$$\begin{aligned}\text{Volume} &= l^3 = (5)^3 \text{ cm}^3 \\ &= 5 \times 5 \times 5 \text{ cm}^3 \\ &= 125 \text{ cm}^3\end{aligned}$$



1. Circle the correct answer in each of the following.

(a) If a cube has a side length of 4 cm, what is its volume?

- A 64 cm^3 B 16 cm^3 C 32 cm^3 D 48 cm^3

(b) A carpenter needs to calculate the volume of a wooden cube with a side length of 10 cm. What is the volume?

- A 100 cm^3 B 1000 cm^3 C 50 cm^3 D 10 cm^3

(c) A jeweler creates a cube-shaped gold piece with a side length of 2 cm. What is its volume?

- A 4 cm^3 B 6 cm^3 C 8 cm^3 D 10 cm^3

(d) A cube-shaped box has a volume of 27 cm^3 . What is the length of one side of the cube?

- A 3 cm B 9 cm C 6 cm D 4 cm

2. A pastry chef is preparing a cube-shaped jelly dessert. If the mold has sides of 10 cm, how much jelly (in cm^3) will the mold hold?

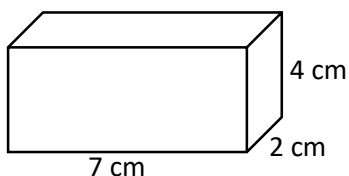
3. An industrial technician is stacking cube-shaped shipping crates, each with a side length of 1.5 meters, in a warehouse. What is the volume of one crate, and how much total space is occupied by 10 such crates?

Lesson 9: Volume of a cuboid

Example

The diagram below shows a cuboid of length 7 cm, breadth 2 cm and height 4 cm.

Find the volume of the cuboid.



Solution

$$l = 7 \text{ cm}, b = 2 \text{ cm and } h = 4 \text{ cm}$$

$$\begin{aligned}\text{Volume of cuboid} &= l \times b \times h \\ &= (7 \times 2 \times 4) \text{ cm}^3 \\ &= 56 \text{ cm}^3\end{aligned}$$

4. Circle the correct answer in each of the following.

(a) What is the formula for calculating the volume of a cuboid?

A $\text{length} \times \text{width} \times \text{height}$ B $\text{length} + \text{width} + \text{height}$ C length^3 D $\text{length} \times \text{width}$

(b) A cuboid-shaped water tank has a length of 10 m, a width of 4 m, and a height of 2 m.

What is its volume?

A 80 m^3 B 40 m^3 C 20 m^3 D 100 m^3

(c) A cuboid has a volume of 120 cm^3 , a length of 10 cm, and a width of 4 cm. What is its height?

A 3 cm B 4 cm C 5 cm D 6 cm

5. A fish tank has a length of 45 cm, a width of 25 cm, and depth of 10 cm. Calculate the volume of the fish tank.

6. The diagram shows a cuboid which has a square base. The height of the cuboid is 4 m and the volume of the cuboid is 36 m^3 . Find the length of the cuboid.

